Feature Engineering, Model Optimization & Performance Comparison

Abstract

This project focuses on predicting housing prices using the California Housing Dataset. The dataset was analyzed and preprocessed using Python libraries such as Pandas, NumPy, and Scikit-Learn. Feature engineering and scaling techniques were applied to improve model performance. Three regression models—Linear Regression, Ridge Regression, and Decision Tree Regressor—were trained and compared using RMSE and R^2 Score. The project demonstrates the importance of data preprocessing and model selection in machine learning.

Introduction

Machine Learning enables systems to learn patterns from data and make predictions. Housing price prediction is a common regression problem where factors such as income, location, population, and house age affect property values. This project aims to build predictive models and compare their performance for accurate housing price estimation.

Objectives

- Analyze the California Housing Dataset.
- Perform data preprocessing and feature engineering.
- Apply feature scaling techniques.
- Train multiple regression models.
- Evaluate and compare model performance.

Dataset Description

The California Housing Dataset contains information related to housing areas in California. Important features include median income, house age, average rooms, average bedrooms, population, occupancy, latitude, and longitude. The target variable is the median house value.

Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-Learn
- Matplotlib

Methodology

1. Loaded the California Housing Dataset.
2. Performed data cleaning and inspection.
3. Applied feature engineering techniques.
4. Used StandardScaler for feature scaling.
5. Split the dataset into training and testing sets.
6. Trained Linear Regression, Ridge Regression, and Decision Tree Regressor models.
7. Evaluated models using RMSE and R^2 Score.
8. Compared the performance of all models.

Results

Model	RMSE	R^2 Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree Regressor	0.7303	0.5930

The model with the lowest RMSE and highest R^2 Score was considered the best-performing model.

Visualization

Graphs were created to understand data distribution and compare model performance. An Actual vs Predicted Values plot was also used to evaluate prediction accuracy.

Conclusion

The project successfully implemented feature engineering and model optimization techniques for housing price prediction. The performance of multiple regression models was compared, demonstrating how preprocessing and appropriate model selection improve machine learning results.

References

1. Scikit-Learn Documentation
2. Python Documentation
3. Pandas Documentation
4. NumPy Documentation
5. California Housing Dataset Documentation